

$S(x) = A^{-1}x$ is the unique function satisfying equations (1) and (2).

~~1800-20~~ From eqn (2), each x in $\mathbb{R}^n$ is the image of at least one vector in $\mathbb{R}^n$. $\therefore$ By Defn, T is onto $\mathbb{R}^n$.

~~1800-20~~ Let, $u, v \in \mathbb{R}^n$ s.t. $T(u) = T(v) \Rightarrow Au = Av$

$\Rightarrow Au + (-Av) = Av + (-Av) \Rightarrow A(u - v) = 0$ (Algebraic property ix)

$\Rightarrow A(u - v) = 0 \Rightarrow A \cdot (-1 \cdot v) = 0$ (Theorem 5(b) from Chapter 1)

$\Rightarrow A(-1 \cdot v) = 0 \Rightarrow A \cdot (-1 \cdot v) = 0$ (Theorem 5(a) from Chapter 1)

$\Rightarrow A(u - v) = 0 \therefore T(u - v) = 0$. By Theorem

Let, $u, v \in \mathbb{R}^n$ s.t. $T(u) = T(v) \Rightarrow Au = Av \Rightarrow A^{-1}(Au) = A^{-1}(Av)$

$\Rightarrow (A^{-1}A)u = (A^{-1}A)v$ (Theorem 2(a)) $\Rightarrow Iu = Iv$ (By the defn. of inverse)

$\Rightarrow \boxed{u = v}$ (By the defn. of identity matrix)

$\therefore$ By defn, T is one-to-one. $\square$

Alternate Method

Given T is an invertible linear transformation. $\therefore$ By Theorem 9, the standard matrix A of T is invertible. $\therefore T: \mathbb{R}^n \rightarrow \mathbb{R}^n$, A is an $n \times n$ matrix. By the Invertible Matrix Theorem and the equivalence of the statements (a), (e) and (i), the linear transformation T is both one-to-one and onto $\mathbb{R}^n$. $\square$

36) Let T be a linear transformation that maps $\mathbb{R}^n$ onto $\mathbb{R}^n$. Show that T^{-1} exists and maps $\mathbb{R}^n$ onto $\mathbb{R}^n$. Is T^{-1} also one-to-one? $\in \mathbb{R}^n$

Solution: By defn, for all $y \in \mathbb{R}^n$, there exists at least one x s.t. $T(x) = y$.

By Theorem 10, from Chapter 1, there exists a unique standard $n \times n$ matrix A s.t. $T(x) = Ax$ for all $x \in \mathbb{R}^n$. By the Invertible Matrix Theorem and the equivalence of the statements (a) and (i), we can conclude that A is invertible. By Theorem 9, T is invertible, and $T^{-1}(x) = A^{-1}x$ is the unique function satisfying the equations $T(T^{-1}(x)) = x$ for all $x \in \mathbb{R}^n$ and $T^{-1}(T(x)) = x$ for all $x \in \mathbb{R}^n$. $T^{-1}(x)$ is also a linear transformation.